

Towhid Ahmed

Generative AI Engineer · Open to remote — US/EU time zones

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Professional Summary

Generative AI Engineer with 1+ year of freelance experience building **production-grade LLM applications**, **multi-agent systems**, and **RAG** solutions for SaaS and e-commerce platforms. Developed 5 AI applications using **LangChain**, **LangGraph**, and **FastAPI**, achieving **35% hallucination reduction**, **25% retrieval relevance improvement**, and **98% test coverage**. Experienced in **RAG pipelines**, **LLMOps**, **Docker**, and **LangSmith**.

Technical Skills

Programming Languages: Python, SQL

Generative AI: LangChain, LangGraph, Advanced RAG, AI Agents, Prompt Engineering

Databases: Pinecone, ChromaDB, PostgreSQL

Backend & Deployment: FastAPI, Docker, Git & GitHub Actions

Testing & Monitoring: Pytest, LangSmith, LLM Guardrails

LLMs & AI Frameworks: Llama, OpenAI API, Groq, Hugging Face Transformers, Sentence Transformers

Software Engineering: Data Structures & Algorithms (Solved 400+ problems), OOP, LLM System Design

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | June 2026 – July 2026

- Enhanced information reliability and user trust by developing a **3-agent LangGraph workflow** with **citation-based RAG**, significantly **reducing AI hallucinations**.
- Powered seamless document processing and interaction by engineering **7 production-ready REST APIs** using **FastAPI**, **GPT-4 Turbo**, and **Pinecone**.
- Enabled **1,000+** document knowledge-base updates without redeployment.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- **Automated customer interactions and improved consistency** using a multi-agent **LangGraph pipeline** (Intent, Retrieval, Response) with Pinecone RAG and **conversational memory**.
- Slashed AI inference costs and maintained high execution speed by integrating **Groq**, Hugging Face embeddings, and **Pinecone Serverless**.
- **Built a scalable support infrastructure** with a Streamlit UI and a FastAPI backend featuring **6+ REST APIs** for knowledge uploads and session management.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, HuggingFace, Tavily GitHub | Demo

- **Designed 4-agent LangGraph workflow** routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily with DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25%** and **reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3**, **BAAI embeddings**, **Pinecone**, **FastAPI**, **Streamlit**, **Docker**; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Created **clinical AI** using **4-agent LangGraph pipeline** (Router+Safety+Retrieval+Answer) with **RAG** and **web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with **JWT** auth.
- Achieved **98/98** passing tests across **guardrails**, **RAG evaluation**, and **API workflows**.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology | 2023-2027

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMOps — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward